# MOSFETs: Electronic Devices and Circuits

Comprehensive Study Report

Based on 24EECE2001 Lecture Material

## Contents


**1 Introduction to MOSFETs** **2**

**2 Device Structure and Physical Operation** **2**
2.1 Physical Structure . . . . . 2
2.2 Operation with Zero Gate Voltage . . . . . 2
2.3 Creating a Channel for Current Flow . . . . . 3

**3 MOSFET Current–Voltage Characteristics** **3**
3.1 Derivation of the Triode Region Relationship . . . . . 3
3.2 Saturation Region and Pinch-Off . . . . . 3
3.3 Channel-Length Modulation . . . . . 4

**4 MOSFET Circuits at DC** **4**
4.1 Worked Examples . . . . . 4

**5 MOSFET Amplifier Design & Small-Signal Models** **5**
5.1 Small-Signal Condition and Transconductance . . . . . 5
5.2 Small-Signal Equivalent Circuits . . . . . 5

**6 Basic MOSFET Amplifier Configurations** **6**
6.1 Common-Source (CS) Amplifier . . . . . 6
6.2 Common-Source with Source Resistance ($R_S$) . . . . . 6
6.3 Common-Gate (CG) Amplifier . . . . . 6
6.4 Common-Drain (CD) Amplifier / Source Follower . . . . . 7

**7 Biasing in MOS Amplifier Circuits** **7**
7.1 Biasing by Fixing $V_{GS}$ . . . . . 7
7.2 Biasing by Fixing $V_G$ and Connecting a Source Resistor ($R_S$) . . . . . 7

# 1 Introduction to MOSFETs

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is the most widely used electronic device, particularly in the design of integrated circuits (ICs). Its fundamental principle involves using the voltage between two terminals (gate and source) to control the current flowing through a third terminal (the drain).

Compared to Bipolar Junction Transistors (BJTs), MOSFETs can be manufactured in a significantly smaller silicon area, their fabrication process is simpler, and their operation requires comparatively little power. These properties allow manufacturers to pack hundreds of millions of MOSFETs onto a single Very-Large-Scale-Integrated (VLSI) memory or microprocessor chip.

MOSFETs are categorized into two major types: **Enhancement-type** and **Depletion-type**. Each type can be fabricated with either an n-channel (NMOS) or a p-channel (PMOS). Complementary MOS (CMOS) technology utilizes both NMOS and PMOS transistors on the same substrate, forming the basis for most modern digital circuitry.

# 2 Device Structure and Physical Operation

## 2.1 Physical Structure

An enhancement-type NMOS transistor is fabricated on a p-type substrate (a single-crystal silicon wafer providing physical support). Two heavily doped n-type regions (indicated as $n^+$) are implanted into the substrate to form the **source (S)** and **drain (D)**.

A very thin layer of silicon dioxide ($SiO_2$), an excellent electrical insulator with thickness $t_{ox}$ (typically 2 to 50 nm), is grown over the surface between the source and drain. A metal or polysilicon layer is deposited on top of this oxide to form the **gate (G)** electrode. The substrate itself forms the fourth terminal, known as the **body (B)**.

Typical dimensions for the channel length ($L$) range from 0.1 to 3 $\mu$m, while the channel width ($W$) typically ranges from 0.2 to 100 $\mu$m.

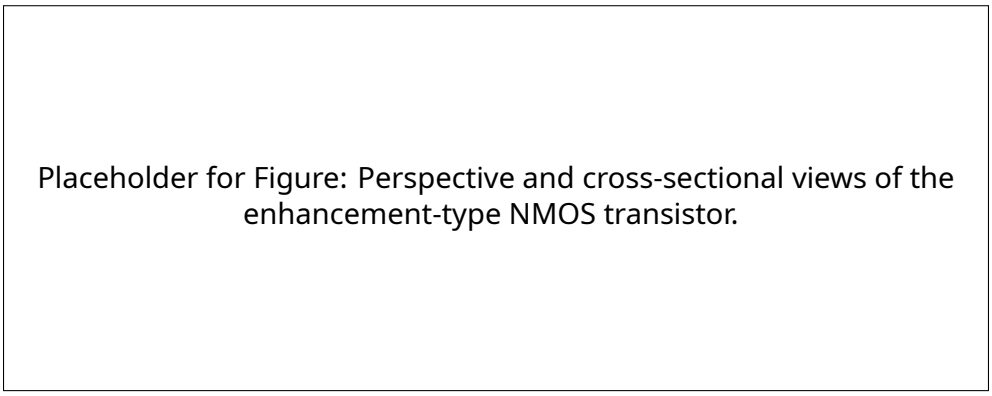



Figure 1: Physical structure of the enhancement-type NMOS transistor

## 2.2 Operation with Zero Gate Voltage

With no bias voltage applied to the gate, two back-to-back pn-junctions exist in series between the drain and the source. These back-to-back diodes prevent current conduction from drain to source when a voltage $V_{DS}$ is applied. The resistance is incredibly high (on the order of $10^{12}\ \Omega$), meaning the device is effectively cut off.

### 2.3 Creating a Channel for Current Flow

To establish a current path, a positive voltage $V_{GS}$ is applied to the gate. The gate and the substrate form a parallel-plate capacitor, with the oxide layer acting as the dielectric. The positive gate voltage pushes mobile holes down into the substrate, leaving behind a depletion region of uncovered bound negative acceptor ions. As $V_{GS}$ increases, it eventually attracts free electrons from the $n^+$ source and drain regions into the region directly beneath the gate.

This accumulation of electrons inverts the p-type substrate surface into an n-type layer, known as an **inversion layer** or induced n-channel. The minimum value of $V_{GS}$ required to form a conducting channel is called the **threshold voltage**, $V_t$ (typically 0.5 V to 1.0 V).

The voltage applied in excess of the threshold is the **overdrive voltage**, defined as:

$$V_{OV} = V_{GS} - V_t$$

Increasing $V_{OV}$ attracts more electrons, lowering the channel resistivity and "enhancing" the channel.

## 3 MOSFET Current–Voltage Characteristics

### 3.1 Derivation of the Triode Region Relationship

Let the capacitance per unit gate area be $C_{ox} = \epsilon_{ox}/t_{ox}$, where $\epsilon_{ox} = 3.9\epsilon_o = 3.45\times10^{-11}$ F/m. The total charge in the channel is proportional to the overdrive voltage. For a small drain-to-source voltage $V_{DS}$, the charge per unit channel length is $C_{ox}W(V_{GS} - V_t)$.

The voltage $V_{DS}$ creates an electric field $|E| = V_{DS}/L$, causing electrons to drift with velocity $v_{drift} = \mu_n|E|$, where $\mu_n$ is the surface mobility of electrons. The drain current is the product of charge density and drift velocity:

$$i_D = \left[(\mu_n C_{ox})\frac{W}{L}(V_{GS} - V_t)\right] v_{DS}$$

Here, $k'_n = \mu_n C_{ox}$ is the **process transconductance parameter** (A/V$^2$), and $W/L$ is the **aspect ratio**. For small $v_{DS}$, the MOSFET acts as a linear resistor $r_{DS}$ controlled by $V_{GS}$.

As $V_{DS}$ becomes larger, the voltage along the channel varies. The average voltage driving the charge becomes $(V_{GS} - V_t - \frac{1}{2}V_{DS})$. Thus, in the **triode region** ($V_{DS} < V_{OV}$):

$$i_D = k'_n\frac{W}{L}\left((V_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2\right)$$

### 3.2 Saturation Region and Pinch-Off

As $V_{DS}$ approaches $V_{OV}$, the channel depth at the drain end tapers to almost zero. When $V_{DS} = V_{GS} - V_t$, the channel is said to be **pinched off**. Increasing $V_{DS}$ beyond this point does not alter the channel shape significantly; the current saturates at its maximum value for that specific $V_{GS}$.

In the **saturation region** ($V_{DS} \geq V_{OV}$):

$$i_D = \frac{1}{2}k'_n\frac{W}{L}(V_{GS} - V_t)^2$$

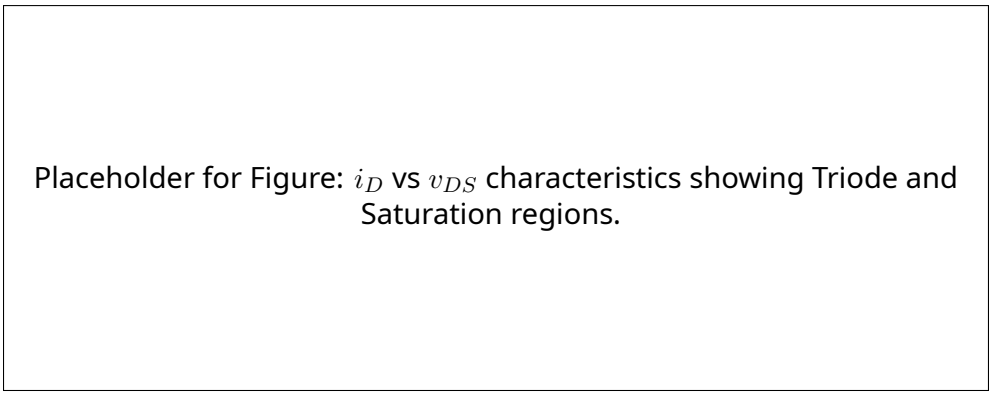



Figure 2: MOSFET Output Characteristics

### 3.3 Channel-Length Modulation

In practical devices, increasing $V_{DS}$ beyond $V_{OV}$ forces the pinch-off point to move slightly toward the source, reducing the effective channel length from $L$ to $L - \Delta L$. Because $i_D$ is inversely proportional to length, current increases slightly with $V_{DS}$. To model this, we introduce the channel-length modulation parameter $\lambda$ (with dimensions $\text{V}^{-1}$):

$$i_D = \frac{1}{2}k'_n \frac{W}{L}(V_{GS} - V_t)^2(1 + \lambda V_{DS})$$

Extrapolating the saturation curves backward yields an intersection at $V_{DS} = -V_A$, where $V_A = 1/\lambda$ is the **Early voltage**. The finite slope implies an output resistance $r_o = V_A/I_D$.

## 4 MOSFET Circuits at DC

To analyze MOSFET DC circuits, we typically assume the transistor operates in the saturation region (if used as an amplifier), calculate currents and voltages, and then verify that $V_{DS} \geq V_{GS} - V_t$.

### 4.1 Worked Examples

**Example 1: DC Design**
*Problem:* Design a circuit to operate an NMOS at $I_D = 0.4$ mA and $V_D = +0.5$ V. The supplies are $V_{DD} = +2.5$ V and $V_{SS} = -2.5$ V. The device has $V_t = 0.7$ V, $L = 1$ $\mu$m, $W = 32$ $\mu$m, and $\mu_n C_{ox} = 100$ $\mu\text{A/V}^2$. Assume $\lambda = 0$.
*Solution:* First, calculate $k_n = \mu_n C_{ox}(W/L) = 100 \times 10^{-6} \times (32/1) = 3.2$ $\text{mA/V}^2$. Using the saturation current equation: $I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2$,

$$0.4 = \frac{1}{2}(3.2)(V_{GS} - 0.7)^2 \implies (V_{GS} - 0.7)^2 = 0.25 \implies V_{GS} = 1.2\ \text{V}$$

Since the gate is tied to ground (0 V), the source voltage is $V_S = -V_{GS} = -1.2$ V. Now calculate $R_S$:

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-1.2 - (-2.5)}{0.4\ \text{mA}} = 3.25\ \text{k}\Omega$$

Calculate $R_D$:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{2.5 - 0.5}{0.4\ \text{mA}} = 5\ \text{k}\Omega$$

Check saturation: $V_{DS} = 0.5 - (-1.2) = 1.7$ V. $V_{OV} = 1.2 - 0.7 = 0.5$ V. Since $1.7 > 0.5$, saturation is verified.

**Example 2: Diode-Connected MOSFET**

*Problem:* Determine $R$ for $I_D = 80\ \mu$A. $V_{DD} = 3$ V, $V_t = 0.6$ V, $\mu_n C_{ox} = 200\ \mu$A/V$^2$, $L = 0.8\ \mu$m, $W = 4\ \mu$m. The drain is tied directly to the gate.

*Solution:* $k_n = 200 \times 10^{-6} \times (4/0.8) = 1$ mA/V$^2$. Since $V_{DG} = 0$, $V_{DS} = V_{GS}$. This implies $V_{DS} > V_{GS} - V_t$, so the device is inherently in saturation.

$$I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2 \implies 0.08 = 0.5 \times 1 \times (V_{GS} - 0.6)^2$$

$(V_{GS} - 0.6)^2 = 0.16 \implies V_{GS} - 0.6 = 0.4 \implies V_{GS} = 1.0$ V. Therefore, $V_D = 1.0$ V. The resistor $R$ is $R = (V_{DD} - V_D)/I_D = (3 - 1.0)/0.08$ mA $= 25$ k$\Omega$.

# 5 MOSFET Amplifier Design & Small-Signal Models

To use the MOSFET as an amplifier, it is biased at a specific quiescent (Q) point in the saturation region. A small time-varying signal $v_{gs}(t)$ is superimposed on the DC bias $V_{GS}$.

## 5.1 Small-Signal Condition and Transconductance

The total instantaneous drain current is:

$$i_D = \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2 = \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2$$

For linear amplification, the nonlinear term must be negligible: $\frac{1}{2} k_n v_{gs}^2 \ll k_n (V_{GS} - V_t) v_{gs}$, which simplifies to $v_{gs} \ll 2V_{OV}$. The linear parameter relating $i_d$ to $v_{gs}$ is the **transconductance**, $g_m$:

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS} = V_{GS}} = k_n (V_{GS} - V_t) = k_n V_{OV}$$

Alternative expressions for $g_m$:

$$g_m = \sqrt{2 I_D k_n' \frac{W}{L}} = \frac{2 I_D}{V_{OV}}$$

These relations show that $g_m$ is proportional to the square root of the bias current $I_D$ and the aspect ratio.

## 5.2 Small-Signal Equivalent Circuits

- **Hybrid-$\pi$ Model:** Consists of an infinite input resistance at the gate, a voltage-controlled current source $g_m v_{gs}$ from drain to source, and an output resistance $r_o = V_A/I_D$ in parallel with the source.
- **T-Model:** Models the transistor with a current source $g_m v_{gs}$ in the drain and an equivalent resistance $1/g_m$ in the source lead, making it highly useful when a physical resistor is connected to the source.

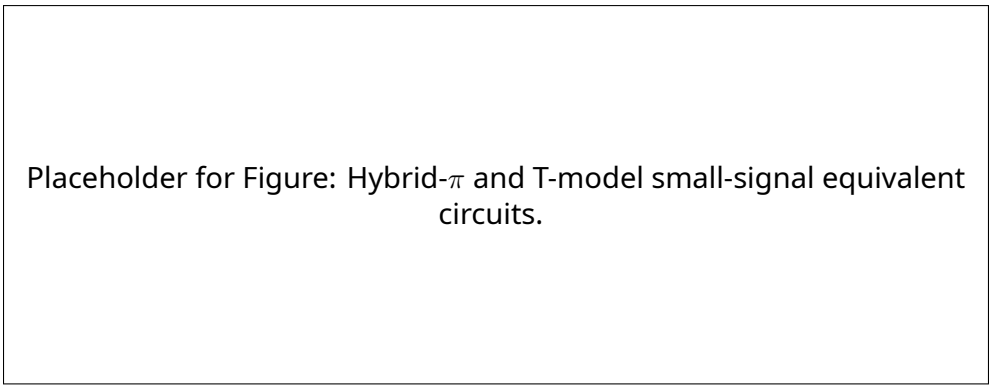



Figure 3: MOSFET Small-Signal Equivalent Circuits

# 6 Basic MOSFET Amplifier Configurations

## 6.1 Common-Source (CS) Amplifier

The CS configuration is the workhorse of MOS amplifiers, typically used to realize the bulk of voltage gain.

- **Input Resistance ($R_{in}$):** $\infty$ (ideal).
- **Output Resistance ($R_o$):** $R_D||r_o$.
- **Voltage Gain ($A_v$):** $A_v = -g_m(R_D||R_L||r_o)$.

While it provides high gain, its output resistance is relatively high.

## 6.2 Common-Source with Source Resistance ($R_S$)

Placing a resistor $R_S$ in the source lead introduces negative feedback (degeneration).

- **Voltage Gain:** Reduces the gain by a factor of $(1 + g_m R_S)$:

$$A_v = -\frac{g_m(R_D||R_L)}{1 + g_m R_S} = -\frac{R_D||R_L}{1/g_m + R_S}$$

This trades lower voltage gain for increased linearity and bias stability.

## 6.3 Common-Gate (CG) Amplifier

The gate is grounded, the input is applied to the source, and the output is taken from the drain.

- **Input Resistance ($R_{in}$):** $1/g_m$ (very low).
- **Voltage Gain:** $A_v = +g_m(R_D||R_L)$ (non-inverting).

Used primarily in high-frequency applications (like the cascode amplifier) because of its superior bandwidth compared to the CS stage.

### 6.4 Common-Drain (CD) Amplifier / Source Follower

The input is applied to the gate, and the output is taken from the source. The drain is connected to $V_{DD}$ (AC ground).

- **Voltage Gain:** Near unity. $A_v = \frac{R_L}{R_L + 1/g_m}$.
- **Output Resistance:** $1/g_m$ (very low).

The CD stage acts as an excellent **voltage buffer**, connecting high-resistance sources to low-resistance loads without loss of signal voltage.

## 7 Biasing in MOS Amplifier Circuits

Establishing a stable, predictable Q-point is essential for preventing the amplifier from clipping (entering cutoff or triode regions) during signal swings.

### 7.1 Biasing by Fixing $V_{GS}$

The most straightforward method is supplying a constant DC voltage $V_{GS}$. However, because the threshold voltage $V_t$ and transconductance parameter $k_n'$ vary widely between devices (even from the same production batch), fixing $V_{GS}$ results in massive variations in $I_D$. This leads to poor Q-point stability and is generally avoided.

### 7.2 Biasing by Fixing $V_G$ and Connecting a Source Resistor ($R_S$)

This is a standard and robust biasing technique. The gate is connected to a fixed voltage $V_G$ (often established via a voltage divider), and a resistor $R_S$ is connected between the source and ground. The gate-source voltage is determined by:

$$V_{GS} = V_G - I_D R_S$$

This setup provides **negative feedback** (degeneration resistance). If for any reason $I_D$ attempts to increase (e.g., due to temperature rise or a device swap), the voltage drop across $R_S$ increases. Since $V_G$ is fixed, this increased source voltage reduces $V_{GS}$, which in turn forces $I_D$ back down toward its original value, stabilizing the bias point effectively.

Placeholder for Figure: Biasing using $V_G$ and a source resistance $R_S$.

Figure 4: Practical Biasing Circuits for MOS Amplifiers